

Question Paper

Exam Date & Time: 13-Jun-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - JUNE 2025

SUBJECT: CHM 1071/CHM_1071 - ENGINEERING CHEMISTRY

(MAKE-UP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) i) Explain the construction and working of calomel electrode. Write any two disadvantages of it. (4)
ii) Write any two differences between galvanic and electrolytic cells.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) i) A galvanic cell consists of iron plate immersed in 1.5 M FeSO_4 solution and copper plate immersed in 15 M solution of CuSO_4 at 298 K. If $E^\circ_{\text{cell}} = 1.56 \text{ V}$. Write the cell reaction, cell representation and calculate the EMF of the cell. (4)
ii) Give reason: Salt bridge is used in the construction of an electrochemical cell.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Give reasons for the following: (2)
i) Oxidant must be free from traces of CO_2 in AFC.
ii) Weston cadmium cell is used for emf measurements.

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2A) i) A solution of Tryptophan has an absorbance at 280 nm of 0.54 in a 0.5 cm length cuvette. Given the absorbance coefficient of tryptophan is $6.4 \times 10^3 \text{ Lmol}^{-1} \text{ cm}^{-1}$. What is the concentration of solution? (4)
ii) Explain the conductometric titration of strong acid Vs strong base with illustration.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) i) Explain the desalination of brackish water using electrodialysis method. (4)
ii) Write any two ways by which priming can be avoided in a boiler.

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 2C) Discuss any two methods of removal of scales. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 3A) Mention any two distinguishing features between the following: (4)
i) Cathodic and anodic protection methods of preventing corrosion.
ii) Galvanization and tinning.

Course Outcome: "CO3" / Cognitive Level: "Understand"

3B) Explain the phenomenon of caustic embrittlement in steel with relevant chemical reactions. Mention two methods of preventing it. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

3C) Explain the effect of the following factors on the rate of corrosion. (2)
i) Nature of corrosion product
ii) Temperature.

Course Outcome: "CO3" / Cognitive Level: "Understand"

4A) i) How does the molecular weight of a polymer affect the glass transition temperature? (4)
ii) A polymer sample has the following composition:
25 molecules of molecular mass each = 45,000
20 molecules of molecular mass each = 20,000
5 molecules of molecular mass each = 25,000
Calculate its number-average, weight-average molecular mass and polydispersity index of the polymer.

Course Outcome: "CO5" / Cognitive Level: "Application"

4B) i) List the classification of liquid crystals based on shape. (4)
ii) Describe the working principle of liquid crystal display in detail.

Course Outcome: "CO5" / Cognitive Level: "Understand"

4C) Explain the role of the matrix and the dispersed phase in composites. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

5A) i) The emf of the cell reaction $\text{Cd} + 2\text{AgCl} \rightarrow \text{CdCl}_2 \cdot 5/2 \text{H}_2\text{O} + 2\text{Ag}$ is 0.6753 V and 0.6915 V at 298 K and 273 K. Calculate ΔH and ΔG at 298 K. (4)
ii) Explain the electroless plating of copper with suitable reactions.

Course Outcome: "CO1" / Cognitive Level: "Application"

5B) i) What are thin films? Write two differences between PVD and CVD techniques of obtaining thin films. (4)
ii) How are polymers classified based on structure and tacticity.

Course Outcome: "CO5" / Cognitive Level: "Understand"

5C) i) Give reason: Corrosion is accelerated by deposition of dust or impurities in the crevices. (2)
ii) Differentiate: Electrochemical series from Galvanic series.

Course Outcome: "CO3" / Cognitive Level: "Understand"

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Event Name			Subject
MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025			CHM 1071-CHM:ENGINEERING CHEMISTRY
#	Course Outcome	Description	Marks

1	1	CO 1	14
2	2	CO 2	4
3	3	CO 3	12
4	4	CO 4	6

* **NA** - CO Not Selected

Event Name	Subject
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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination -
Jun 2025

CHM 1071-CHM:ENGINEERING
CHEMISTRY

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	10	30(60%)
3	Application	5	20(40%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)